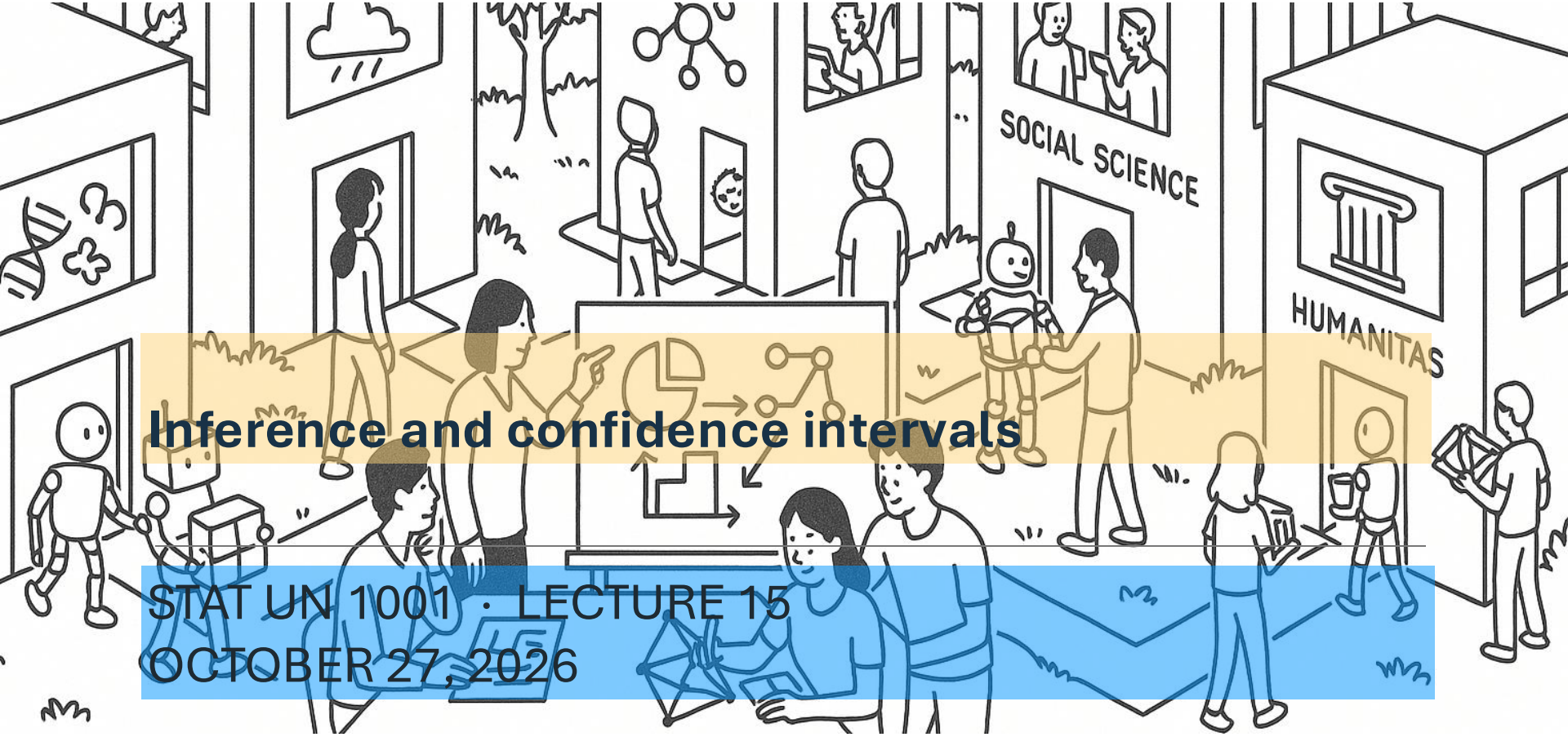


Inference and confidence intervals

STAT UN 1001 · LECTURE 15

OCTOBER 27, 2026

Today's plan

1	**Mini-project 2 checkpoint**	
2	**Quiz 3 questions**	
3	**Sampling distributions**	Why $\hat{p}$ changes from one random sample to another.
4	**Inference and confidence intervals**	How a sample gives evidence about an unknown population parameter.
5	**Hypothesis-testing logic**	A first look at claims, evidence, and decision errors.

A sample proportion is a random variable

Population parameter p	A fixed, usually unknown proportion in the population.
Chance process	Randomly select a sample of n people from that population.
Outcome: $\hat{p}$	Compute the proportion in the selected sample with the trait.
Random variable: $\hat{p}$	A different random sample can produce a different $\hat{p}$.

Before the sample, $\hat{p}$ can vary.
After the sample, $\hat{p}$ is one observed statistic.

One population, many possible samples

Population

Suppose the true proportion of adults who support a policy is $p = 0.88$. That value **does not change** when we take another sample.

Repeated chance process

Randomly sample 1,000 adults, calculate $\hat{p}$, put the sample back conceptually, and repeat.

Possible outcomes

The first sample might give $\hat{p} = 0.873$; another might give 0.887; a third might give 0.901. The values change because the samples differ.

Which quantity stays fixed across the repetitions: p or $\hat{p}$? Why?

The sampling distribution

For repeated random samples of the same size, the sampling distribution describes the possible values of $\hat{p}$ and how likely they are.

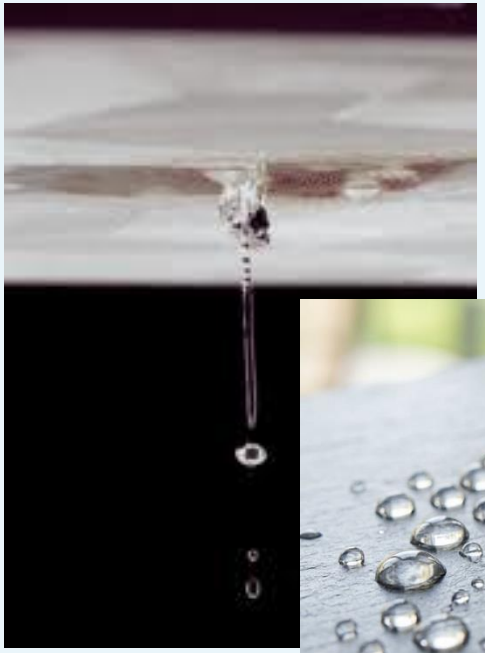
Where do $\hat{p}$ values center?	Near the true population proportion p.
How much do $\hat{p}$ values vary?	The standard error describes a typical sample-to-sample distance.
When can we use a normal model?	When the sampling assumptions and success/failure conditions are reasonable.

Inference uses one observed $\hat{p}$ to reason about the unknown p.

The sampling distribution



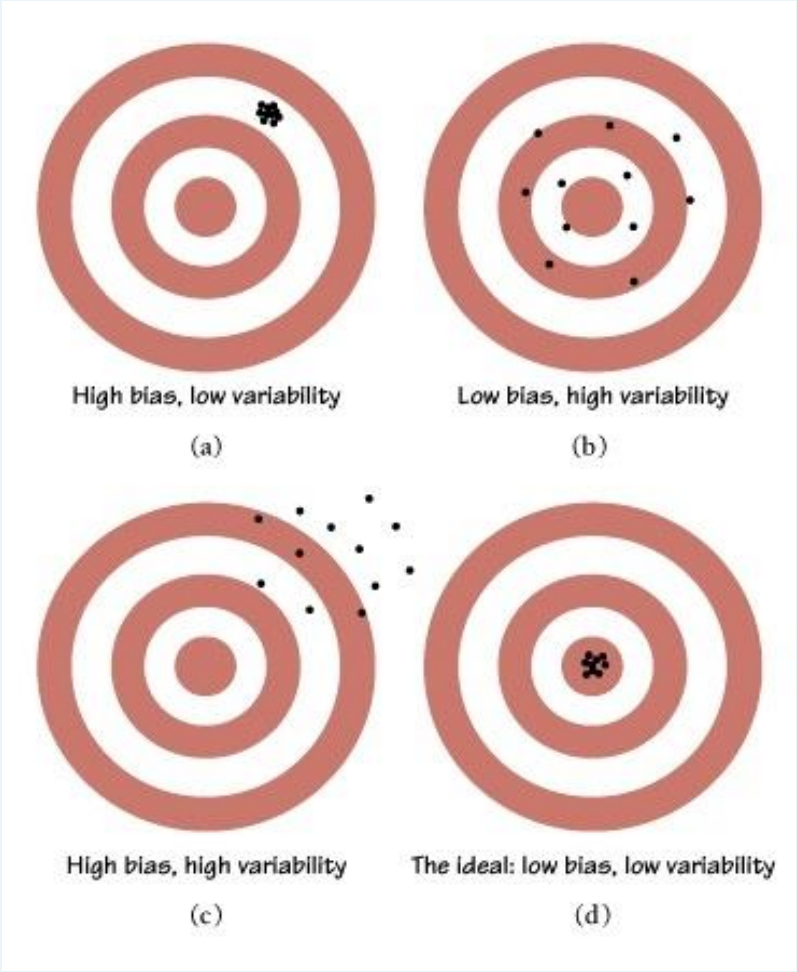
Parameter P



Sampling Distribution of $\hat{p}$



The sampling distribution



From sample to population

Population question	What proportion p of all adults supports the policy?
Observed data	In one random sample of 1,000 adults, 887 support it. Thus $\hat{p} = 0.887$.
Modeling idea	A random sample gives a $\hat{p}$ near p, with ordinary sample-to-sample variation.
Inference question	Which population values of p are plausible given $\hat{p} = 0.887$ and the model?

A confidence interval gives a plausible range for the population parameter.

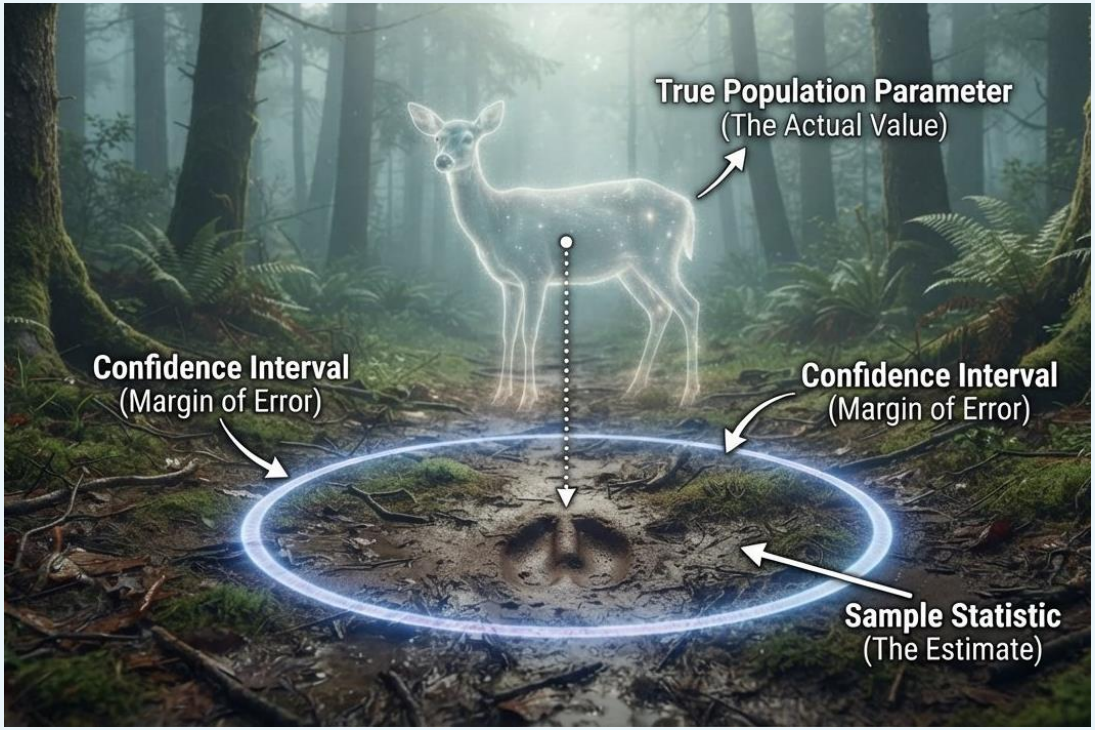
Confidence interval ingredients

confidence interval = point estimate ± margin of error

The **margin of error** combines a critical value and the standard error of the point estimate.

Point estimate	**$\hat{p} = 0.887$**	what the sample says
Standard error	**SE = 0.010**	typical variation
95% critical value	**$z^* = 1.96$**	confidence level

Confidence Interval – Inferential THINKING



What decide the range of inference?

- Center/Starting point
  -> Estimate
- Confidence
  -> Critical Values
- Model and Assumptions
  -> Standard Error

Critical Values

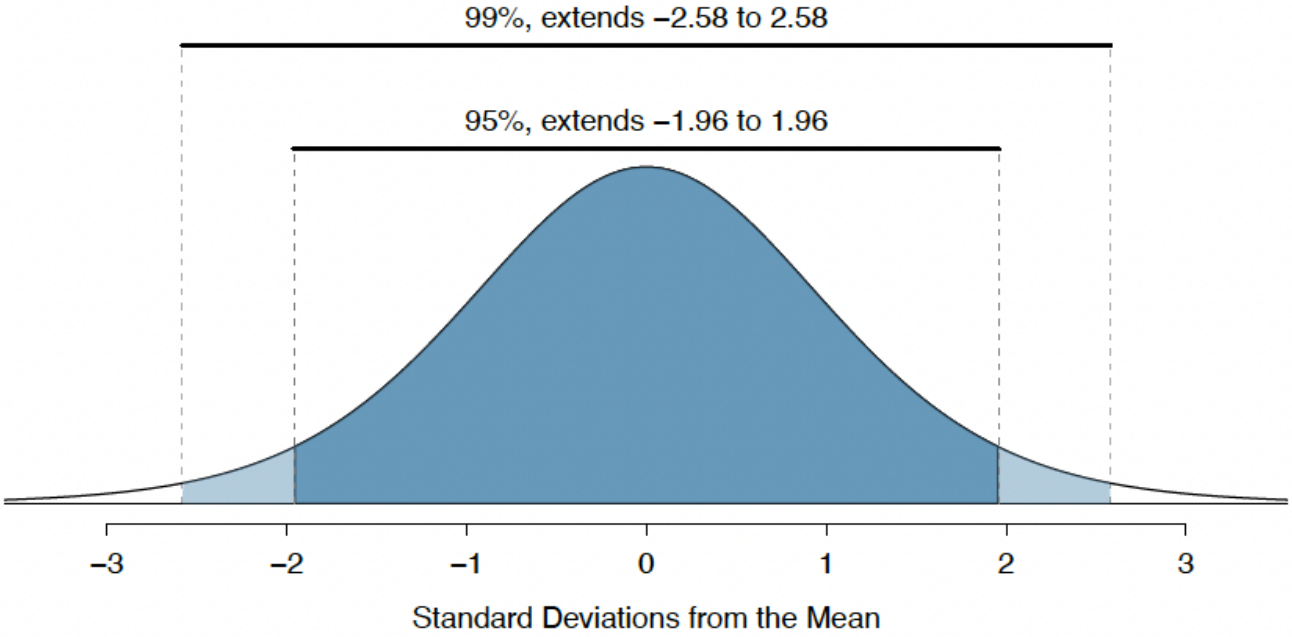
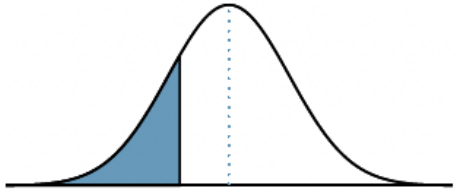


Figure 5.7: The area between -$z^\star$ and $z^\star$ increases as $z^\star$ becomes larger. If the confidence level is 99%, we choose $z^\star$ such that 99% of a normal normal distribution is between -$z^\star$ and $z^\star$, which corresponds to 0.5% in the lower tail and 0.5% in the upper tail: $z^\star = 2.58$.

Critical Values

Find the critical value corresponding to 90% confidence.



Second decimal place of Z										
0.09	0.08	0.07	0.06	0.05	0.04	0.03	0.02	0.01	0.00	Z
0.0002	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	−3.4
0.0003	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0005	0.0005	0.0005	−3.3
0.0005	0.0005	0.0005	0.0006	0.0006	0.0006	0.0006	0.0006	0.0007	0.0007	−3.2
0.0007	0.0007	0.0008	0.0008	0.0008	0.0008	0.0009	0.0009	0.0009	0.0010	−3.1
0.0010	0.0010	0.0011	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013	0.0013	−3.0
0.0014	0.0014	0.0015	0.0015	0.0016	0.0016	0.0017	0.0018	0.0018	0.0019	−2.9
0.0019	0.0020	0.0021	0.0021	0.0022	0.0023	0.0023	0.0024	0.0025	0.0026	−2.8
0.0026	0.0027	0.0028	0.0029	0.0030	0.0031	0.0032	0.0033	0.0034	0.0035	−2.7
0.0036	0.0037	0.0038	0.0039	0.0040	0.0041	0.0043	0.0044	0.0045	0.0047	−2.6
0.0048	0.0049	0.0051	0.0052	0.0054	0.0055	0.0057	0.0059	0.0060	0.0062	−2.5
0.0064	0.0066	0.0068	0.0069	0.0071	0.0073	0.0075	0.0078	0.0080	0.0082	−2.4
0.0084	0.0087	0.0089	0.0091	0.0094	0.0096	0.0099	0.0102	0.0104	0.0107	−2.3
0.0110	0.0113	0.0116	0.0119	0.0122	0.0125	0.0129	0.0132	0.0136	0.0139	−2.2
0.0143	0.0146	0.0150	0.0154	0.0158	0.0162	0.0166	0.0170	0.0174	0.0179	−2.1
0.0183	0.0188	0.0192	0.0197	0.0202	0.0207	0.0212	0.0217	0.0222	0.0228	−2.0
0.0233	0.0239	0.0244	0.0250	0.0256	0.0262	0.0268	0.0274	0.0281	0.0287	−1.9
0.0294	0.0301	0.0307	0.0314	0.0322	0.0329	0.0336	0.0344	0.0351	0.0359	−1.8
0.0367	0.0375	0.0384	0.0392	0.0401	0.0409	0.0418	0.0427	0.0436	0.0446	−1.7
0.0455	0.0465	0.0475	0.0485	0.0495	0.0505	0.0516	0.0526	0.0537	0.0548	−1.6
0.0559	0.0571	0.0582	0.0594	0.0606	0.0618	0.0630	0.0643	0.0655	0.0668	−1.5

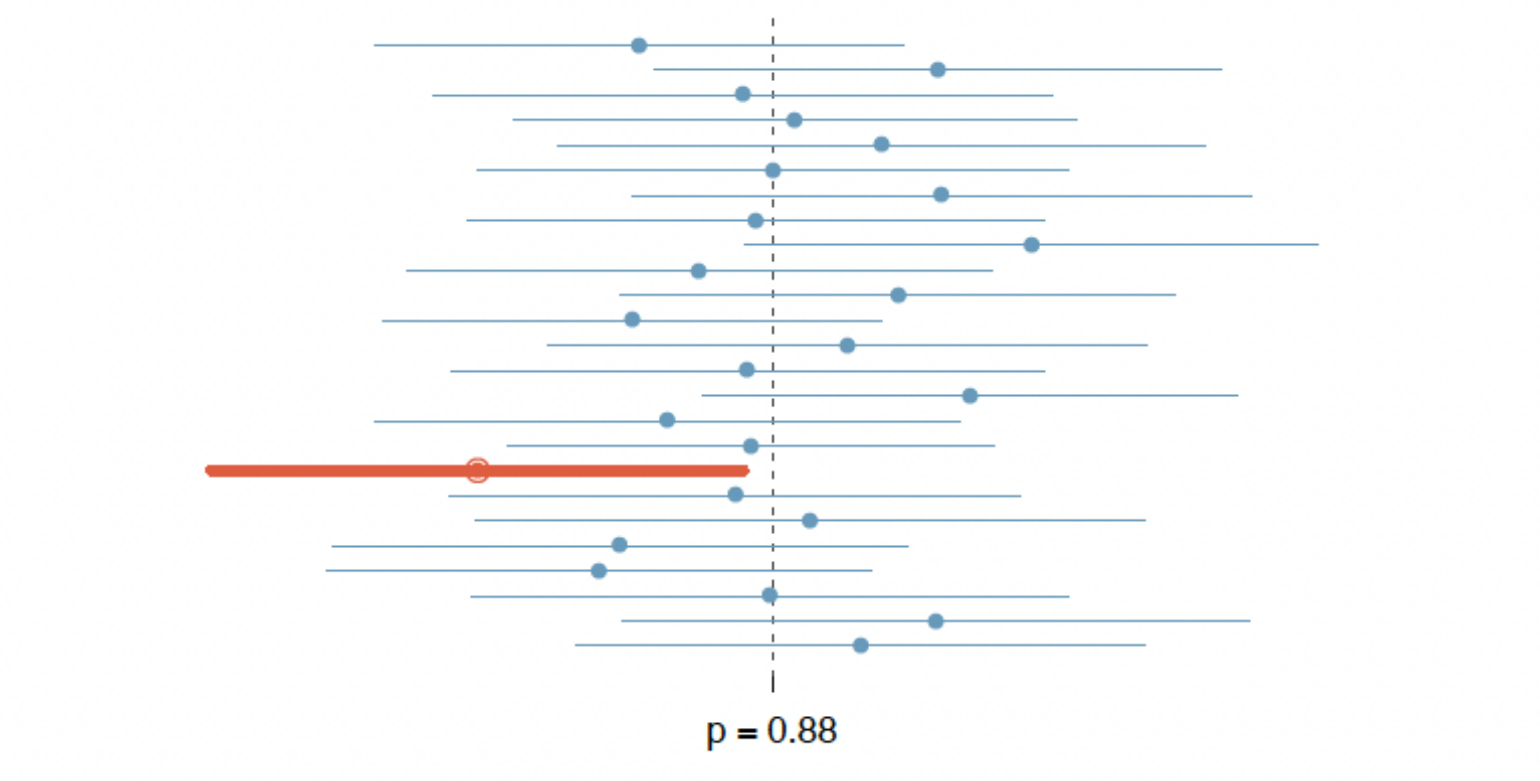
Figure 5.6: what repeats and what stays fixed

The population parameter p	Stays fixed. The population has not changed.
The sample proportion $\hat{p}$	Changes because each random sample differs.
The 95% confidence interval	Changes because it is built from that sample's $\hat{p}$ and standard error.

Of 25 intervals in the book, 24 captured p and 1 missed. That long-run performance defines the 95% method.

Quick check: What does the number “95%” describe?

Figure 5.6: what repeats and what stays fixed



What “95% confident” means

Careful wording

If we repeat the sampling process many times and build intervals the same way, about 95% of those intervals contain the true population parameter.

Avoid

“95% of the data lie in the interval.” The interval concerns a population parameter, not individual data.

Avoid

“There is a 95% probability p is in this finished interval.” In this setup p is fixed.

Use

“This method captures p about 95% of the time.”

Hypothesis testing asks a different question

Confidence interval question

Which population values are plausible, given our sample and a model for sampling variability?

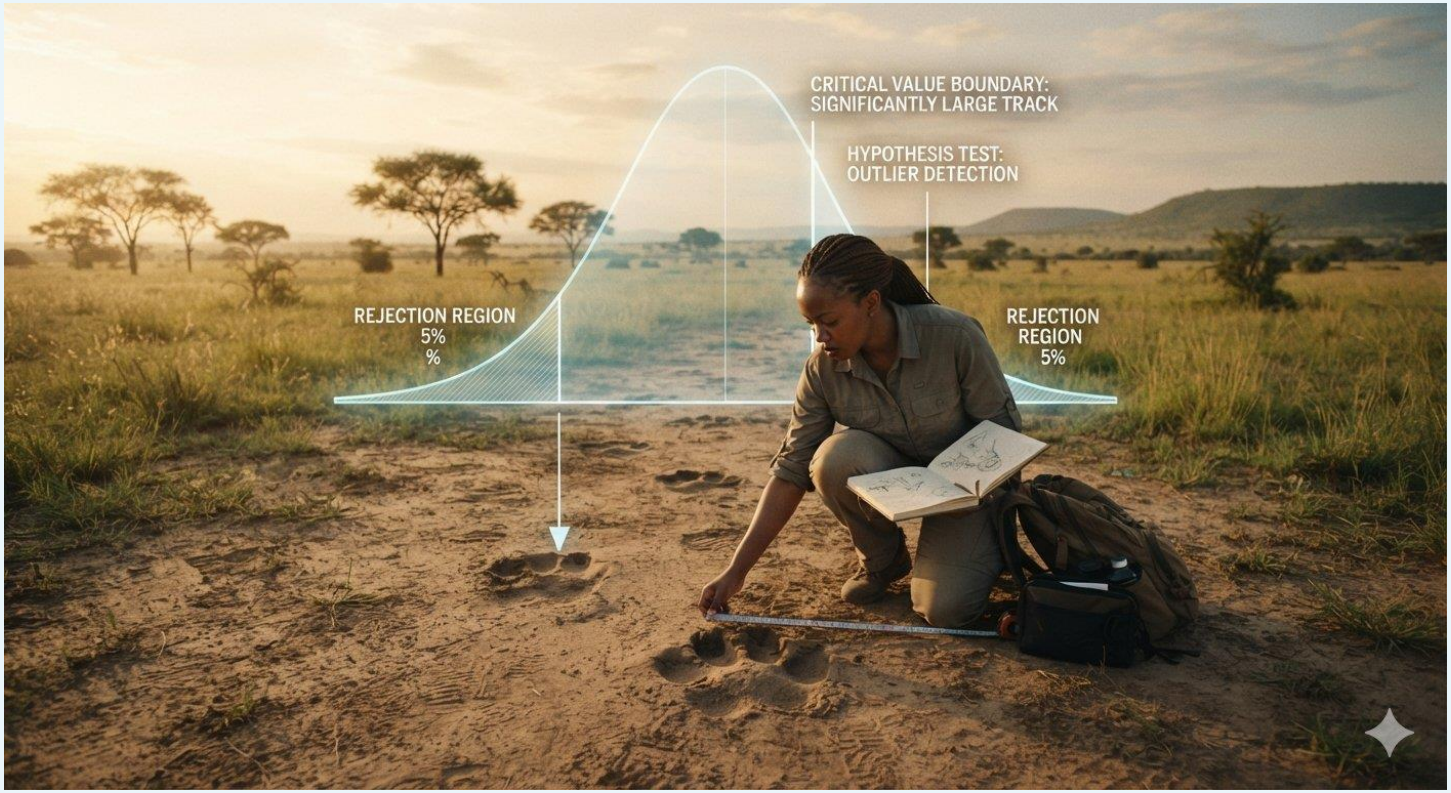
Hypothesis-testing question

If a stated population claim were true, would a sample result like ours count as unusually strong evidence against that claim?

Core pieces

The null hypothesis states the chance-model claim. The alternative states the competing claim. Data supply evidence, not certainty.

Hypothesis testing - Inferential THINKING



Hypothesis-testing definitions

Term	Definition	Example: a report claims p = 0.90
Null hypothesis (H_0)	The claim we test, usually “no effect” or a stated value	H_0: p = 0.90
Alternative (H_a)	The competing claim the data might support	H_a: p ≠ 0.90
Test statistic	How far the sample result sits from H_0, in standard errors	z = (0.887 − 0.90) / 0.0095 ≈ −1.37
p-value	Chance of a result at least this extreme if H_0 were true	2 × P(Z ≤ −1.37) ≈ 0.17
Significance level (α)	Evidence cutoff, chosen before seeing the data	α = 0.05
Conclusion	Reject H_0 if p-value ≤ α; otherwise fail to reject	0.17 > 0.05: fail to reject; 90% stays plausible

Decision errors: one useful analogy

Decision	Reality	Name
Reject H_0	H_0 was true	Type I error
Fail to reject H_0	H_0 was false	Type II error
Either decision	The corresponding statement was true	Correct decision

Courtroom: H_0 means innocent.
Type I: convict an innocent person. Type II: fail to convict a guilty person.